Materials Science
M.Sc. education program

crei.skoltech.ru/cest/education/materials-science-program

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Where to get information

Where to look:

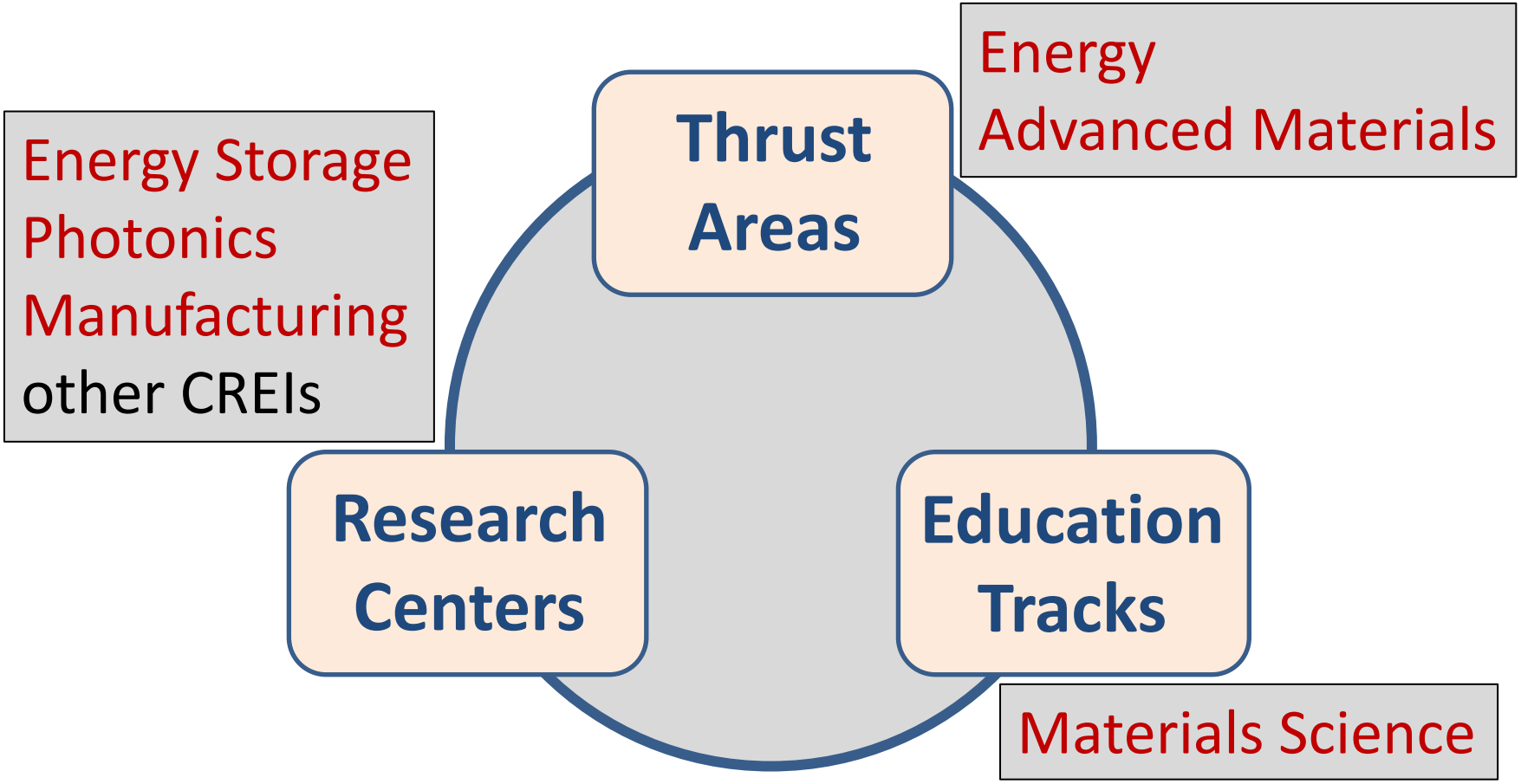
- crei.skoltech.ru/cest/education/materials-science-program

Whom to ask:

- Your academic/research advisor
- Program Coordinator – Andriy Zhugayevych
  - Experimental Materials Science – Artem Abakumov
  - Computational Materials Science – Andriy Zhugayevych

Skoltech research & education environment

- No separate theory or experiment departments
- No separate physics or chemistry departments



Materials Science track: courses

Core courses

- Survey of Materials **(Term 1B)**
- Materials Chemistry **(Term 2)**

Experimental Materials Science

Recommended courses

- Introduction to Solid State Physics**(2)**
- Material Structure Characterization Methods **(3)**
- Organic Materials for Electronics, Energy Conversion and Storage **(3)**
- Electrochemistry **(4)**
- Carbon Nanomaterials **(4)**

Computational Materials Science

Recommended courses

- Computational Chemistry and Materials Modeling **(2)**
- Structure and Property of Materials **(3)**
- Fundamentals of Device Physics **(4)**
- Advanced Solid State Physics **(6)**

Elective courses

- discuss with your advisor

Materials Science track: other activities

Understand your load:

- 1 full course = 20 hours per week (except of Term 1B)
- *Hint*: combine research and education

Seminars (1 hour per week):

- Research Seminar "Advanced Materials Science" – core
- Energy Colloquium – optional
- Computational Materials Science seminar – for subtrack
- Your group seminars – per advisor

Background courses (half-course per term):

- Math for Engineers, Quantum Mechanics (Term 1B!)
- English, Writing, Presentation, Pedagogy

M.Sc. Research in Materials Science

In brief

- *Highly interdisciplinary*
- Materials for energy storage – Abakumov, Antipov, Stevenson
- Materials for optoelectronics – Nasibulin, Troshin
- Computational Materials Science
  – Oganov, Perebeinos, Shapeev, Tretiak, Zhugayevych
- Theoretical Materials Science – Buchachenko, Fine, Skvortsov
- *… with details on faculty web-pages: example*
- Research topics will be covered on Survey of Materials course

M.Sc. Thesis 2016-2018

- Materials for energy conversion and storage (11 students)
- Materials for optoelectronics (3 students)
- Other materials (2 students)

Academic calendar and important deadlines

Academic Year - MSc YEAR 1

| 27-Aug-18 | 03-Sep-18 | 10-Sep-18 | 17-Sep-18 | 24-Sep-18 | 01-Oct-18 | 08-Oct-18 | 15-Oct-18 | 22-Oct-18 | 29-Oct-18 | 05-Nov-18 | 12-Nov-18 | 19-Nov-18 | 26-Nov-18 | 03-Dec-18 | 10-Dec-18 | 17-Dec-18 | | |

Experimental Materials Science: courses

Experimental Materials Science

Recommended courses

- Introduction to Solid State Physics **(2)**
- Material Structure Characterization Methods **(3)**
- Organic Materials for Electronics, Energy Conversion and Storage **(3)**
- Electrochemistry **(4)**
- Carbon Nanomaterials **(4)**

Elective courses

- discuss with your advisor

Computational Materials Science: courses

Computational Materials Science

Recommended courses

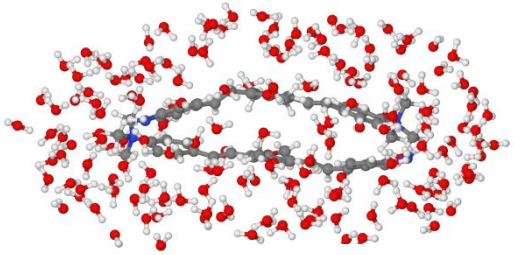
- Computational Chemistry and Materials Modeling **(2)**
- Structure and Property of Materials **(3)**
- Fundamentals of Device Physics **(4)**
- Advanced Solid State Physics **(6)**

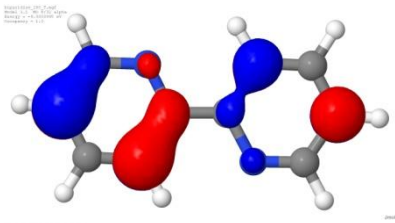
Elective courses

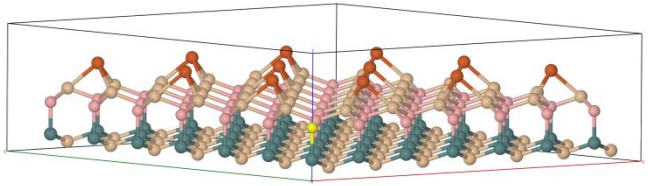
- discuss with your advisor

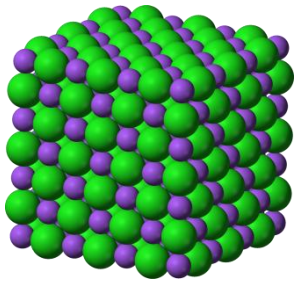
Computational Chemistry & Materials Modeling

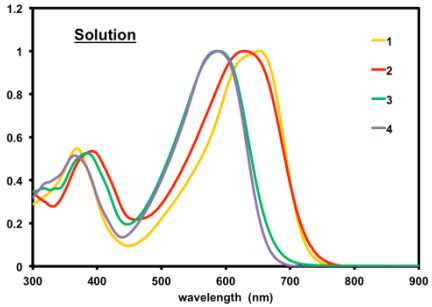
(Sergei Tretiak and Andriy Zhugayevych, Term 2 or 6)











- Whom to attend
  - Everybody who study materials at nanoscale or from first principles, theoretical or experimental
- Learning outcomes (for “beginners”)
  - Minimum knowledge to read scientific articles
  - Minimum set of skills to start first principle materials modeling at any research institution

Other recommended courses

- **Structure and Property of Materials** (Artem Oganov, Term 3 or 7)
  - This course is an introductory subject in the field of materials science and crystallography. The goal is to introduce students to basic concepts of structure-property relations for materials at the microscopic level.
- **Fundamentals of Device Physics** (Vasili Perebeinos, Term 4)
  - The course provides a graduate level overview of physical principles of electronic and optoelectronic devices.
- **Advanced Solid State Physics** (Boris Fine, Term 6 or 2)
  - The course presents an overview of solid-state physics with emphasis on the quantum properties of solids. It covers quantum theory of electronic and lattice degrees of freedom, magnetism and superconductivity, including, in particular, strongly-correlated electronic systems and high-temperature superconductivity.